

JAVA EXCEPTION HANDLING

Input/Output Exception (IOException)

Experiment	Input/Output Exception Handling
Class Name	InputOutputDemo
Exception	IOException
Language	Java

1. What is an Input/Output Exception?

IOException is a Java exception that occurs when an input or output operation fails or is interrupted. It is commonly associated with files, streams, keyboard input, network input/output, and other data-transfer operations. **IOException** is a checked exception, so Java normally requires it to be handled using **try-catch** or declared using **throws**.

2. Objective

To understand and demonstrate how an **IOException** can be handled in Java using the **try-catch** mechanism and display a suitable message when an input/output error occurs.

3. Task Description

Create a Java program named **InputOutputDemo**. Inside the main method, deliberately throw an **IOException** with an explanatory message. Catch the exception using a catch block and print a user-friendly message indicating that an input/output error occurred.

4. Step-wise Execution

Step 1: The program starts execution from the `main()` method.

Step 2: A try block is entered.

Step 3: An **IOException** object is explicitly created and thrown using the `throw` statement.

Step 4: Java stops normal execution of the try block and searches for a matching catch block.

Step 5: The catch block catches the **IOException** object in the variable `e`.

Step 6: The program displays the input/output error message.

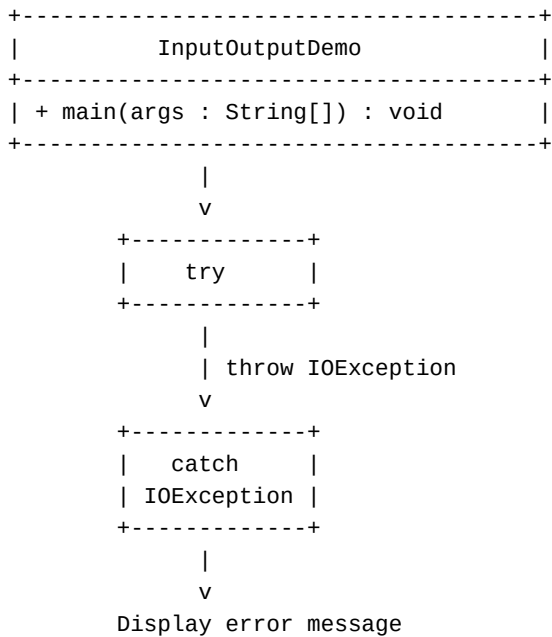
Step 7: The program ends normally after handling the exception.

5. Algorithm

1. Start.
2. Enter the `main()` method.
3. Begin a try block.
4. Create and throw an **IOException** with the message "Input or output error".
5. Catch the **IOException** in the catch block.
6. Display "Exception: Input/Output error occurred".

7. Stop.

6. UML Diagram



7. Program Code

```
import java.io.IOException;

class InputOutputDemo {
    public static void main(String[] args) {
        try {
            throw new IOException("Input or output error");
        } catch (IOException e) {
            System.out.println("Exception: Input/Output error occurred");
        }
    }
}
```

8. Code Explanation

import java.io.IOException; — Imports the IOException class from Java's java.io package.

class InputOutputDemo — Defines a class named InputOutputDemo.

public static void main(String[] args) — Defines the main method, which is the starting point of the program.

try — Contains code that may produce an exception.

throw new IOException(...) — Creates and throws an IOException object with a message.

catch (IOException e) — Catches the IOException and stores the exception object in e.

System.out.println(...) — Displays a message indicating that an input/output error occurred.

9. Expected Output / Output Screenshot

```
Exception: Input/Output error occurred
```

The output shows the message printed after the IOException is caught by the catch block.